

Homework IoT – Exercise 2

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1 BLE Beacons Logic

As clarified in the assignment parameters, the goal of the system is to track the prostheses so that patients do not lose them when they are not wearing them, rather than tracking the patients themselves.

The optimized logic relies on detecting the state of the denture (worn vs. not worn) by monitoring its temperature. The temperature inside the oral cavity is generally around 36-37°C, while the ambient temperature in a nursing home rarely exceeds 25-28°C.

- We establish a temperature threshold $T_{th} = 32^{\circ}\text{C}$.
- If the measured temperature is $\geq T_{th}$, we assume the denture is **worn**. In this state, the BLE beacon is not transmitted, and the microcontroller enters a deep sleep mode, waking up infrequently (e.g., every 60 seconds) solely to sample the temperature.
- If the temperature falls below T_{th} , the denture is **not worn**. The device then begins transmitting BLE advertising packets periodically (every 10 seconds) to allow the gateways to localize it.

2 Additional Sensors

To implement this logic, a **temperature sensor** is required. Since the space inside a dental prosthesis is extremely limited, the optimal solution is not to add a discrete external sensor, but to utilize the **on-chip temperature sensor** integrated into almost all modern BLE SoCs. This avoids increasing the Bill of Materials (BOM) and saves critical physical space.

3 Improved Firmware Pseudocode

```
1 #define TEMP_THRESHOLD 32.0          // Threshold in Celsius
2 #define SLEEP_WORN_INTERVAL 60      // Sleep duration if worn (
3 #define BEACON_INTERVAL 10         // Beacon interval if not worn (
4                                     seconds)
5 void setup() {
6     init_ble();
7     init_temp_sensor();
8 }
9
10 void loop() {
11     // 1. Read temperature from the internal sensor
12     float current_temp = read_internal_temperature();
13 }
```

```

14 // 2. Decision logic
15 if (current_temp < TEMP_THRESHOLD) {
16     // Denture is outside the mouth: transmit for localization
17     send_BLE_beacon();
18     go_to_deep_sleep(BEACON_INTERVAL);
19 } else {
20     // Denture is worn: no transmission needed
21     go_to_deep_sleep(SLEEP_WORN_INTERVAL);
22 }
23 }

```

4 Commercial Components

The proposed sensing logic can be implemented using standard ultra-low-power IoT components:

- **BLE SoC: Nordic Semiconductor nRF52832.** This is an industry-standard chip for space-constrained applications, available in a WLCSP package (approx. 3.0 x 3.2 mm). It features an integrated temperature sensor with 0.25°C resolution and draws roughly 50 μ A for a single measurement, making it ideal for this use case.
- **Battery: CR1216 or CR1220** (3V lithium coin cells, 25-35 mAh capacity). Alternatively, silver-oxide batteries (e.g., SR626SW) could be used if the thickness constraints of the prosthesis are particularly strict.

5 Energy Saving Estimate

To estimate the energy savings, we formalize the assumptions based on a realistic daily usage pattern:

- **Usage Time:** The patient wears the prosthesis for 16 hours a day and removes it for 8 hours (during the night or if misplaced).
- **Beacon Interval:** 10 seconds (as given in the original firmware).
- **Energy Cost:** We assume the energy spent on checking the internal temperature is negligible compared to the energy required for a BLE radio transmission (E_{tx}).

Original Firmware (Continuous Beacons):

The device transmits one packet every 10 seconds, 24 hours a day.

$$N_{tx_orig} = \frac{24 \times 3600}{10} = 8640 \text{ transmissions/day}$$

Improved Firmware:

- **Worn Phase (16 hours):** 0 BLE transmissions. The chip wakes up every 60 seconds for a thermal check (radio remains off).
- **Not Worn Phase (8 hours):** Transmits every 10 seconds.

$$N_{tx_improved} = \frac{8 \times 3600}{10} = 2880 \text{ transmissions/day}$$

Energy Saving Calculation:

The daily energy saving related to radio transmissions is:

$$\text{Savings} = \frac{N_{tx_orig} - N_{tx_improved}}{N_{tx_orig}} \times 100$$

$$\text{Savings} = \frac{8640 - 2880}{8640} \times 100 = 66.6\%$$

Conclusion:

By eliminating unnecessary transmissions while the denture is in the mouth (approximately two-thirds of the day), the active radio energy consumption is reduced by **66.6%**. This effectively triples the expected battery lifetime compared to the original continuous-beaconing solution.